# Statistical Modelling & Analysis (STAT2004)
# Applied Class (Week 7)

*We will go through Questions 2 & 3 together in the Applied Class. Try the other questions yourself during the week and discuss your results with your tutor next week.*

1. Before the launch of the space shuttle *Challenger*, information was available from 23 previous shuttle launches. For launch $i$, let $Y_i = 1$ if an incident of damage to the field-joint O-rings was observed and let $Y_i = 0$ otherwise. Let $x_i$ denote the launch temperature in degrees Fahrenheit. Consider the logistic regression model

$$Y_i \sim \text{Bernoulli}(p_i), \qquad \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_i.$$

The data and model were entered in R as follows.

```
orings <- data.frame(
  temperature = c(53, 57, 58, 63, 66, 67, 67, 67, 68, 69,
                  70, 70, 70, 70, 72, 73, 75, 75, 76, 76,
                  78, 79, 81),
  damage = c(1, 1, 1, 1, 0, 0, 0, 0, 0, 0,
             1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
)

fit <- glm(damage ~ temperature, family = binomial,
           data = orings)
summary(fit)
vcov(fit)
```

Relevant output is shown below.

```
Call:
glm(formula = damage ~ temperature, family = binomial,
    data = orings)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.0611  -0.7613  -0.3783   0.4524   2.2175

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   15.0429     7.3786   2.039   0.0415
 temperature  -0.2322     0.1082  -2.145   0.0320

(Dispersion parameter for binomial family taken to be 1)

> vcov(fit)
              (Intercept) temperature
(Intercept)     54.444183   -0.796385
 temperature    -0.796385    0.011715
```

Let $T_{50}$ be the launch temperature at which the model gives a probability of 0.5 that an O-ring damage incident occurs.

(a) Express $T_{50}$ in terms of $\beta_0$ and $\beta_1$. Hence, determine its maximum likelihood estimate and state the property of maximum likelihood estimators used.

(b) Suppose that $\widehat{\boldsymbol{\theta}}$ is an MLE of $\boldsymbol{\theta} \in \mathbb{R}^d$ and that $g : \mathbb{R}^d \longrightarrow \mathbb{R}$ is differentiable. Recall that

$$g(\widehat{\boldsymbol{\theta}}) \mathrel{\dot\sim} N\Big(g(\boldsymbol{\theta}), \nabla g(\boldsymbol{\theta})^\top \mathcal{I}(\widehat{\boldsymbol{\theta}})^{-1} \nabla g(\boldsymbol{\theta})\Big),$$

where $\mathcal{I}(\boldsymbol{\theta})$ is the expected information matrix. Use this result and the output from `vcov(fit)` to calculate the estimated standard error of $\widehat{T}_{50}$.

(c) Suppose temperature is instead measured in degrees Celsius, with

$$C = \frac{5}{9}(T - 32).$$

Express the logistic regression model in terms of $C$. Determine how the fitted intercept, fitted slope, $\widehat{T}_{50}$ and its estimated standard error change. Explain why the fitted probabilities do not change.

2. The `kyphosis` data frame contains information on 81 children who underwent corrective spinal surgery. The response records whether kyphosis was present or absent following surgery. The explanatory variables are the child's age in months (`Age`), the number of vertebrae involved in the operation (`Number`), and the number of the first, or topmost, vertebra operated on (`Start`). Define a successful outcome to mean that kyphosis was absent following surgery.
The following logistic regression model was fitted in R.

```
data(kyphosis, package = "rpart")
kyphosis$success <- as.integer(kyphosis$Kyphosis == "absent")
fit <- glm(success ~ Age + Number + Start,
           family = binomial, data = kyphosis)
summary(fit)
```

Relevant output is shown below.

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   2.03693    1.44958   1.405  0.16003
Age          -0.01093    0.00645  -1.696  0.08996
Number       -0.41060    0.22486  -1.826  0.06791
Start         0.20651    0.06771   3.050  0.00229

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 83.234  on 80  degrees of freedom
Residual deviance: 61.380  on 77  degrees of freedom
AIC: 69.380
```

Let $p_i$ denote the probability of a successful outcome for child $i$.

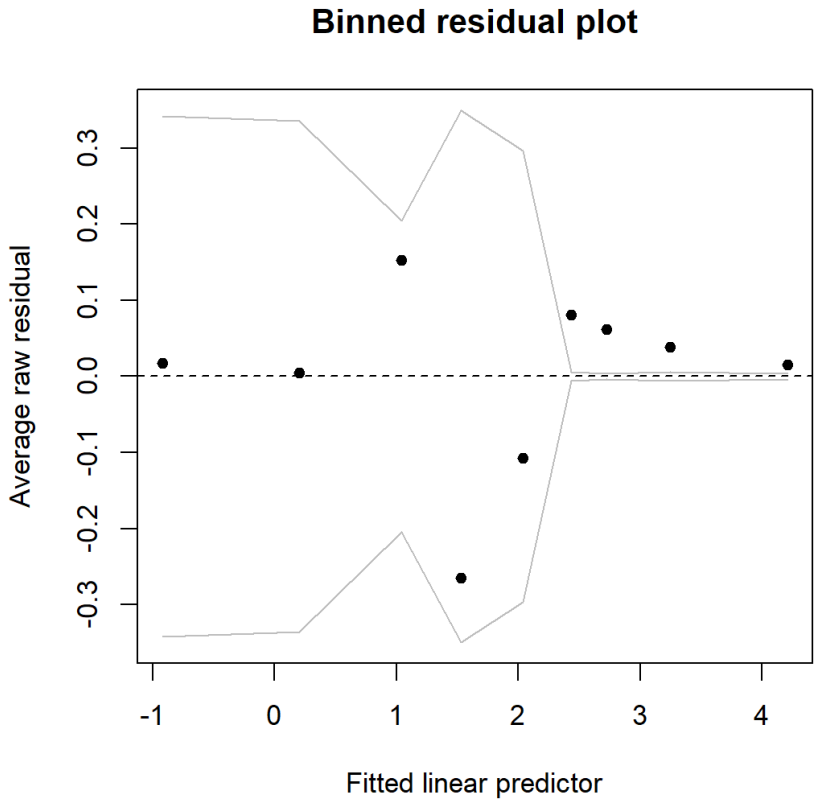
(a) Write down the fitted logistic regression model.

(b) Interpret the fitted coefficient for `Start` in terms of the odds of a successful outcome.

(c) Use the fitted coefficient for `Age` to determine the multiplicative change in the odds of a successful outcome associated with an increase in age of one year, holding `Number` and `Start` fixed. Interpret the result.

(d) Explain why a plot of the individual raw residuals $y_i - \widehat{p}_i$ against the fitted linear predictors is difficult to interpret for this model. How does binning the residuals make the plot more informative?

(e) A binned residual plot for the fitted model is shown below. The grey lines are calculated as approximately two standard errors above and below zero, where the standard error in each bin is estimated using the sample standard deviation of the raw residuals in that bin. For some bins at the right of the plot, the observed responses are nearly all the same. The raw residuals in such a bin consequently have very little empirical variation, causing the grey reference limits to collapse towards zero. Points outside these very narrow limits should therefore not automatically be interpreted as evidence of lack of fit. What does the plot suggest about the adequacy of the fitted model? Justify your conclusion by considering the overall pattern of the binned residuals, as well as the behaviour of the grey reference limits.




The plot was produced using the following code. This code will be useful in Question 3.

```
library(arm)
binnedplot(x = predict(fit, type = "link"),
           y = residuals(fit, type = "response"),
           nclass = 9,
           xlab = "Fitted linear predictor",
           ylab = "Average raw residual")
```

(f) Estimate the probability of a successful outcome for a child aged 96 months when six vertebrae are involved in the operation and the first vertebra operated on is vertebra 8.

(g) Suppose the response were instead coded as 1 when postoperative kyphosis is present. Explain how the fitted regression coefficients and fitted probabilities would change.

3. The `Wells` data contain information from 3020 households in Bangladesh whose wells had unsafe arsenic levels. The response `switch` records whether a household switched to another well. The explanatory variables considered here are distance to the nearest known safe well, arsenic concentration in the household's original well, and years of education of the household head. Load

and prepare the data as follows.

```
data(Wells, package = "carData")
Wells$switch01 <- as.integer(Wells$switch == "yes")
Wells$dist100 <- Wells$distance/100
Wells$educ4 <- Wells$education/4
```

Thus, `dist100` measures distance in hundreds of metres and `educ4` measures education in units of four years.

(a) Fit a logistic regression model for switching wells using `dist100`, `arsenic`, and `educ4` as explanatory variables. Store the fitted model as `fit.wells` and examine the fitted coefficients.

(b) Interpret the estimated coefficient of `dist100` in terms of the odds of switching wells.

(c) Interpret the estimated coefficient of `arsenic` in terms of the odds of switching wells.

(d) Using the code in Question 2(e) as a guide, produce a binned residual plot of the raw residuals against the fitted linear predictors. Use 40 bins. Does this plot provide clear evidence of lack of fit? Explain.

(e) Produce a binned residual plot of the raw residuals against `arsenic`, again using 40 bins. Interpret the plot.

(f) What aspect of the fitted model should be reconsidered in light of the plot in part (e)?